

CHAPTER 4

DEEP-SEA BIOLOGY

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

The shallow water that surrounds land masses accounts for less than 10% of the area of the world's oceans. This region, plus the offshore waters lit by the sun, composes that portion of the world's oceans where most of the primary production takes place. If considered on a volumetric basis, probably 90% of the ocean is composed of dark, cold waters: the deep sea. In this domain no photosynthesis takes place and nearly all of the food consumed there is produced elsewhere. Nevertheless, bacterial chemosynthesis supplies the energetic needs of a few widely scattered and highly specialized communities.

This chapter is divided into six sections. The first, ZONATION, briefly reviews the habitats of the deep sea, while the next section, SAMPLING THE DEEP SEA, discusses how this realm is studied. The third section, ENVIRONMENTAL CHARACTERISTICS, is a précis of the abiotic factors encountered in the deep sea. The last three sections cover ADAPTIONS OF DEEP-SEA ORGANISMS, COMMUNITY ECOLOGY OF THE BENTHOS (the longest section of the three), and MIDWATER COMMUNITY ECOLOGY, respectively.

I. Zonation

There is probably little that one can say to excite students into understanding the abstraction of dividing an area up into named regions for the purposes of accurate communication. This problem is exacerbated by the fact that workers in the field cannot seem to settle on one set of terms and be done with it. (See also Chapter 1, particularly Fig. 1.25.)

abyssal	deep sea	mesopelagic
abyssalpelagic	dysphotic zone	photic zone
aphotic zone	epipelagic	twilight
bathyal	hadal	
bathypelagic	hadalpelagic	

II. Sampling the Deep Sea

Accurate sampling is one of the most difficult aspects of studying the deep sea.

acoustical pinger	<i>Deep Rover</i>	midwater trawl
<i>Alvin</i>	deep-sea cameras	multicorers
anchor dredge	epibenthic sled	net avoidance
box core	fouling the gear	<i>Sea Cliff</i>

1. Sampling the deep-sea midwater or benthic regions requires a very long cable to which specialized sampling gear is tethered. The amount of line needed is two to three times the vertical depth to be sampled. Such long-line work is fraught with dangers and difficulties. Sampling failures may occur when the samplers do not make sound contact with the bottom or when the gear becomes fouled on the way up or down.
2. Other problems confound research in this field. For example, most dredges are not designed to accurately sample the benthos in a quantitative way. In addition, in midwater sampling, many fish are able to avoid the net.
3. New technologies have helped to overcome some of these problems. This equipment includes deep-sea cameras and various submersibles, including remote-operated vehicles (ROVs) and vehicles capable of carrying investigators to great depths. In addition, acoustical devices aid long-line operators by providing accurate information on the location of their gear while they are sampling the deep sea.

III. Environmental Characteristics

Unlike those of the shallow-water or upper-water communities, the abiotic characteristics of the deep sea remain relatively constant for long periods of time. This section is divided into seven subsections, each of which covers a different abiotic factor:

- Light
- Pressure
- Salinity
- Temperature
- Oxygen
- Food
- Biomass

allochthonous	chitinous exoskeletons	oxygen minimum zone
angler fishes	DOM	pressure-dependent physiology
autochthonous	fecal pellets	Santa Barbara Basin
black smoker chimney	float-and-wait predators	thermocline
Carioca Trench	homeoviscous adaptation	wood (cellulose)
chemosynthesis	marine snow	

1. Light from the sun is, for all practical purposes, absent from the deep sea, and, of course, photosynthesis is therefore not possible in these waters.
2. Pressure increases one atmosphere for every 10 meters (≈ 33 feet) of depth. Thus, within the depths encountered in the deep sea, the pressure experienced by its inhabitants ranges from 20 to $> 1,000$ atmospheres. While these pressures seem extraordinary to us, they represent just the opposite condition to the deep-sea organisms; that is, they are ordinary. Both simple measures of growth and sophisticated enzyme analyses indicate that these high pressures are required for normal metabolic activity of deep-sea animals. This phenomenon, called pressure-dependent physiology, extends to Michaelis–Menton kinetics (activation energies and catalytic efficiencies), the biochemistry of lipid membranes (homeoviscous adaptation), and predation strategies (float and wait).
3. Most of the deep sea is more or less uniform in salinity and temperature. Aside from the oxygen minimum zone (ca. 500–1000 m) and a few regional exceptions (for example, the Carioca Trench and the Santa Barbara Basin), most of the deep sea is well oxygenated. Organisms that inhabit the regions of low oxygen concentrations are, of course, adapted to this condition, and they either use less oxygen than other organisms or are very good at extracting it from the water. Oxygen-rich cold water descending from the Arctic and Antarctic regions keeps the deeper waters well supplied with oxygen.
4. Nearly all of the food (carbon/energy, or C/E, sources) used in the deep sea is produced elsewhere and then transported into the region. (That is, the food has allochthonous origins.) Some energy sources are not suitable for direct consumption (for example, fecal pellets and chitinous exoskeletons). This material, as well as marine snow, is decomposed by bacteria, which in turn are consumed by higher trophic levels. Important exceptions to the rule that all food is transported into the deep sea occur in the hydrothermal-vent communities. (N.B.: The C/E sources for these communities are termed autochthonous.)

In hydrothermal-vent communities, chemolithotrophy (chemosynthesis) replaces photosynthesis as the base of the food chain, and the diversity and abundance of organisms are unusually high. In general, over vast regions of the deep-sea benthos, low food levels are responsible for a paucity of deep-sea marine organisms.

IV. Adaptations of Deep-Sea Organisms

The adaptations of deep-sea organisms are forced by the abiotic factors discussed in the previous section and the biotic factors of avoiding predators while securing sufficient food and a mate. This section is divided into two subsections, an untitled one that reviews assorted morphological adaptations and a titled one that explores bioluminescence in the deep sea:

- Unnamed section that considers morphological adaptations of deep-sea organisms
- Bioluminescence in the Deep Sea

abyssal gigantism	isopod	ostracod
amphipod	lantern fish	phosphorescent sea
angler fish	light organs	photophores
<i>Architeuthis</i>	luciferase	Red-Queen Hypothesis
barbel	luciferin	respiration
bioluminescence	luminescent	rhodopsin
copepod	lure	ROVs
countershading	male parasitism	silhouetting
dimorphic eyes	monster camera	squid
hinged jaw	mysids	transparent
Histioteuthidae	olfaction	tubular eyes
ink	ooze	twilight vision

1. While countershading is absent in deep-sea animals, color is still an important feature in mesopelagic organisms. Mesopelagic fish are generally silvery gray to deep black. Purple and bright red are colors common to invertebrates at this depth. These colors absorb the blue bioluminescent light produced by predators, thereby reducing the probability of detection.
2. Several different adaptations of the eyes are seen in deep-sea organisms. Those living at $< 2,000$ m possess large eyes, while those living at the greatest depths ($> 2,000$ m) have small eyes or no eyes at all. Bottom-dwellers often have no eyes, too. Tubular and dimorphic eyes are two other anatomical modifications present in these organisms. Twilight vision refers to an enhancement of the photoreceptive capacity of certain fish by the presence of the vision pigment rhodopsin and an increased density of rods in the retina.
3. Food is scarce in the deep sea, and the organisms that inhabit this world are adapted to consume almost anything. For example, their mouths are much larger than would otherwise be expected, their teeth point backward into the throat, and their body parts may be modified as bioluminescence lures.
4. Male parasitism appears to be another concession to food scarcity and the resulting low population density. Most deep-sea fishes are small, but one of the enigmatic characteristics of deep-sea invertebrates is the phenomenon of abyssal gigantism. The text briefly reviews the hypotheses regarding this characteristic. Abyssal gigantism may be an example of the Red-Queen Hypothesis: Evolutionarily speaking, one must run very fast just to maintain one's position with respect to others in the race. In this case, animals grow larger than their neighbors, who otherwise might become their predators. Nevertheless, abyssal giants are rare; most benthic invertebrates are smaller than similar species present in shallow waters. They also tend to possess simpler body organizations.
5. Bioluminescence in deep-sea organisms is a dominant theme, judging from the number of species possessing photophores and the complexity of those organs. Light production probably serves a variety of functions, including preventing the production of silhouettes, luring prey, and enabling species-specific recognition for mating.

One of the most unusual suggestions regarding bioluminescence is the burglar-alarm hypothesis. According to this theory, a flash of light from the prey confuses the would-be predator enough that it drops (or loses) its meal. Of course, the flash of light might signal other predators in the area that food is available. In that case, both prey and the original predator ought to beat a hasty retreat. See the following references for more information:

- Abrahams, M.V., and L.D. Townsend. 1993. Bioluminescence in dinoflagellates: a test of the burglar alarm hypothesis. *Ecology* 74: 258–260.
- Burkenroad, M.D. 1943. A possible function of bioluminescence. *J. Mar. Res.* 5: 161–164.
- Buskey, E., L. Mills, and E. Swift. 1983. The effect of dinoflagellate bioluminescence on the behavior of a marine copepod. *Limnol. Oceanogr.* 28: 575–579.
- Esaias, W.E., and H.C. Curl, Jr. 1972. Effect of dinoflagellate bioluminescence on copepod ingestion rates. *Limnol. Oceanogr.* 17: 901–906.

34 Instructor's Manual for Marine Biology

Fleisher, K.J., and F.F. Case. 1995. Cephalopod predation facilitated by dinoflagellate luminescence. *Biol. Bull.* 189: 263–272.

Rees, J.F., et al. 1998. The origins of marine bioluminescence: turning oxygen defence mechanisms into deep-sea communication tools. *J. Exp. Biol.* 201: 1211–1221.

V. Community Ecology of the Benthos

It is not easy to study this remote place, but deep submersibles, remote-operated cameras, and grab samples have provided a good picture of the deep-sea benthic environment. This part of the chapter is divided into six sections:

- Faunal Composition
- Community Structure
- Diversity
- Seamounts
- Life History Patterns
- Hydrothermal-Vent and Cold-Seep Communities

abyssal oozes	Florida Escarpment	patchiness
<i>Alvin</i>	food generalists	planktotrophic larvae
angler fish	food specialists	<i>Riftia</i>
anoxic	Galápagos Rift Vent	Rose Garden
area hypothesis	H ₂ S	seamounts
<i>Bathymodiolus</i>	hadal zone	semelparous
black smoker chimneys	hydrocarbon seeps	small-scale disturbance
brine pools	hydrothermal vents	spaghetti worms
<i>Calyptogenia</i>	hypersaline brines	species diversity
CH ₄	infauna	springs
chemolithoautotrophic bacteria	Juan de Fuca Vent	Stability-time hypothesis
chemosynthetic bacteria	life history	sulfate-reducing bacteria
cold seeps	Marianas Back Arc Basin	suspension feeders
competitive exclusion	megafauna	trenches
Cropper theory	methanogenic bacteria	trophic structure
detritus feeders	microhabitats	vent communities
Disturbance theory	North Fiji Basin	vestmentiferan worms
epifauna	oases	whale falls
fermentative bacteria	opportunistic predator	white smokers

1. Although their absolute abundance varies from group to group, all major animal taxa are present in the benthos of the deep sea. Unlike the members of shallow-water communities, few deep-sea animals are suspension feeders. The low density of the deep-sea benthos led to the conclusion that this region is similar to terrestrial deserts, but this view was changed when intensive sampling demonstrated that species diversity is actually quite high in the deep sea.

Corliss, B., and K.A. Silva. 1993. Rapid growth of deep-sea benthic foraminifera. *Geology* 21: 991–995.

Dybas, C.L. 1996. The deep-sea floor rivals rain forests in diversity of life. *Smithsonian* 26(10): 96–106.

Dybas, C.L. 1999. Undertakers of the deep. *Natural History* 108(9): 40–47.

Rex, M.A. 1993. Global-scale latitudinal patterns of species diversity in the deep-sea benthos. *Nature* 365: 636–639.

Rex, M.A. 2000. Latitudinal gradients of species richness in the deep-sea benthos of the North Atlantic. *PNAS* 97: 4082–4085.

2. Three competing ideas have been proposed to explain the relatively high species diversity of the deep-sea benthos. They are known as the Stability-time hypothesis; the Cropper or Disturbance

theory; and area hypothesis. However, none of these ideas in itself may be adequate to explain the high level of species diversity.

Instructors may wish to examine the use of the words “theory” and “hypothesis” in science. Is either one used to imply a different level of sophistication or a different level of credibility? You might have your students explore these terms to determine whether they should really be used interchangeably. Other terms that might be discussed are “law,” “model,” and “paradigm.” This topic is a good place to explore connections between the philosophy of science and the practical use of concepts by scientists.

3. There is an unusual relationship between species diversity and depth in the deep-sea benthos. Although there is considerable variability in the data presented in analyses, it does appear that species diversity increases with depth until the range of 1,500 to 3,000 m. At that point, diversity levels off and then declines. Therefore, diversity-depth plots resemble parabolas. (Refer to Figure 4.41 in the textbook.)
4. The deep-sea benthos is a patchy environment, which is evident from the photographs and observations that have been made. An interesting type of patch is the seamount. Seamounts represent a fascinating departure from the norm of the deep sea. Found in all oceans, these structures rise ≥ 1 km above the seafloor, and yet they are often hundreds to thousands of meters below the surface. Although seamounts are not rare—estimates suggest that some 30,000 or more exist—very few have been studied. These mountains have their own endemic fauna, and therefore the concept of island biogeography can be brought into your discussions of them. Moreover, seamounts obstruct ocean circulation, thus increasing their importance.
5. One unusual type of patch is the vent habitat. Hydrothermal and other vents (seeps) are a most extraordinary and interesting community type. Discovered in 1977, the Galápagos Rift Vent (GRV) has opened a new and marvelous world for study. At some 2,700 m in depth, the GRV possesses a rich and bizarre array of animals that live in warm water (8–16°C) that is rich in reduced sulfur (H_2S).

Baker, E.T. 1991. Megaplumes. *Oceanus* 34(4): 84–91.

Haymon, R.M., and K.C. Macdonald. 1985. The geology of deep-sea hot springs. *Am. Sci.* 73: 441–449.

Lutz, R.A., and R.M. Haymon. 1994. Rebirth of a deep-sea vent. *Nat. Geo.* 186(5): 114–126.

Mullineaux, L.S., P.H. Wiebe, and E.T. Baker. 1991. Hydrothermal vent plumes: larval highways in the deep-sea. *Oceanus* 34(3): 64–68.

Tiercelin, J.-J., et al. 1993. Hydrothermal vents in Lake Tanganyika, East African Rift system. *Geology* 21: 499–503.

Tivey, M.K. 1991. Hydrothermal vent systems. *Oceanus* 34(4): 68–74.

Tunnicliffe, V. 1992. Hydrothermal-vent communities of the deep-sea. *Am. Sci.* 80(4): 336–349.

Ugur, A., et al. 2003. Radiometric dating of sediment cores from a hydrothermal vent zone off Milos Island in the Aegean Sea. *Sci. Total Environ.* 307: 203–215.

6. Several other vents have been discovered as well, including some cold, hypersaline brines and hydrocarbon seeps. All of these communities depend on the energy provided by chemolithotrophic (chemosynthetic) bacteria, which oxidize reduced sulfur and, in the process, fix carbon dioxide.

Aharon, P., H.H. Roberts, and R. Snelling. 1992. Submarine venting of brines in the Gulf of Mexico: observations and geochemistry. *Geology* 20(6): 483–486.

Aloisi, G., et al. CH_4 -consuming microorganisms and the formation of carbonate crusts at cold seeps. *Earth Planet. Sci. Lett.* 203: 195–204.

Campbell, K.A., and D.J. Bottjer. 1993. Fossil cold seeps. *Res. Explor.* 9(3): 326–343.

Sager, W.W., I.R. MacDonald, and R. Hou. 2003. Geophysical signatures of mud mounds at hydrocarbon seeps on the Louisiana continental slope, northern Gulf of Mexico. *Marine Geology* 198: 97–133.

Sahling, H., et al. 2003. Depth-related structure and ecological significance of cold-seep communities—a case study from the Sea of Okhotsk. *Deep Sea Res. Part I: Oceanogr. Res. Pap.* 50: 1391–1410.

Some animals of these habitats filter the bacteria from the water for their food, while others are involved in intimate symbiotic relationships with these prokaryotes. (See Chapter 10 for further information on symbiotic relationships.) Methane-based metabolisms are also known to exist. While spectacular in their own right, vents are even more interesting because they are relatively short lived. Evidence suggests that they last only in the range of decades and that their species composition may change dramatically during that time. This condition raises many questions about their colonization, development, and death. Do the same sorts of ecological principles apply to these habitats as to, say, old field succession or rocky coasts?

- Although rare (ca. one per 25 km²), the large biomass of dead whales provides another type of patch on the landscape of the deep-sea benthos. Here, too, the bacterial biomass (whose C/E source is the sulfide-rich whalebone) is sufficient to support a community similar to that of vent systems.

Schuller, D., D. Kadko, and C.R. Smith. 2004. Use of 210Pb/226Ra disequilibria in the dating of deep-sea whale falls. *Earth Planet. Sci. Lett.* 218: 277–290.

VI. Midwater Community Ecology

Relatively little is known about the animal life of the midwater habitat. This section is divided into four parts:

- Deep Scattering Layers and Vertical Migration
- Species Composition and Distribution
- Food and Feeding
- Life-History Patterns of Midwater Organisms

deep scattering layer (DSL)	krill	resource partitioning
diel vertical migration	lantern fish	sergestid shrimp
euphausiids	midwater	siphonophores
generalist feeders	opportunistic feeders	sonar
interzonal fauna	pasaphaeid shrimp	species composition

- Common animals comprising DSLs include lantern fish, krill, shrimps, squids, and deep-floating siphonophores (phylum Cnidaria). While we can identify the organisms that inhabit this region, we really do not know much about their ecology.
- Although food is scarce in the deep sea, the animals that live there are still selective in their choices; in general, food resources are partitioned. One way this partitioning is accomplished is through differential feeding with depth; another is apparently via specific anatomical differences that permit prey of only certain sizes to be taken.

ENRICHMENT EXERCISES

Deep-Sea Niches

The broad patches of the deep-sea benthos, each dominated by a few species, are vaguely reminiscent of the grassland prairies of midcontinental North America before European colonization. For example, in the east, big bluestem dominated, while further west, big bluestem was the major grass. Trees were found in hollows and along streams and rivers. Of course, scattered throughout these regions, one could find significant numbers of other prairie plants as well.

Although the old prairies are all but gone, the plant species found in them are still widely distributed. Recent studies have demonstrated that the genetic diversity of prairies as a whole is quite high. The species in each locale possess a unique genetic composition (ecotypes). This finding, of course, has been determined by restriction endonuclease cleavage of DNA, followed by a genetic analysis of the DNA fragment size, using gel

electrophoresis. This condition is probably related to the specific microclimatic and edaphic conditions of each habitat.

The following questions may stimulate class discussion: Is the genetic composition of the deep-sea benthos elaborated along lines similar to that found in the prairies? Why or why not? Formulate this idea into a logical series of hypotheses, and devise an experimental protocol to test each hypothesis.

Deep-Sea Biology

The following are some eclectic references that may be of interest to students:

- Angel, M.V., and M. Sibert. (eds.). 1990. Deep sea biology. *Prog. in Oceanogr.* 24(1–4): 1–343.
- Bailey, D.M., and I.G. Priede. 2002. Predicting fish behaviour in response to abyssal food falls. *Mar. Bio.* 141: 831–841.
- Corliss, J.B., and R.O. Ballard. 1977. Oases of life in the cold abyss. *Nat. Geo.* 152(4): 441–453.
- Cronin, T.M. 1997. Orbital forcing of deep-sea benthic species diversity. *Nature* 385: 624–627.
- Etter, R.J., and J.F. Grassel. 1992. Patterns of species diversity in the deep sea as a function of sediment particle size distribution. *Nature* 360: 576–578.
- Grassle, J.F. 1991. Deep-sea benthic diversity. *BioScience* 41(7): 464–469.
- Levin, L.A., et al. 2001. Environmental influences on regional deep-sea species diversity. *Ann. Rev. Eco. Sys.* 32: 51–94.
- Livermore, B. 1993. Bottoms up: satellite images turn up the focus on the ocean's floor. *Sea Frontiers* 39(3): 40–45.
- Prager, E.J., and S.A. Earle. 2000. *The oceans*. New York: McGraw-Hill.
- Rex, M.A. 1997. An oblique slant on deep-sea biodiversity. *Nature* 385: 577–578.
- Rex, M.A., C.T. Stuart, and R.R. Hessler. 1993. Global-scale latitudinal patterns of species diversity in the deep-sea benthos. *Nature* 365: 636–639.
- Snelgrove, P.V.R. 1995. The deep sea: desert and rainforest: debunking the desert analogy. *Oceanus* 38: 25–28.
- Vecchione, M., et al. 2001. Worldwide observations of remarkable deep-sea squids. *Science* 294: 2505–2507.
- Yayanos, A.A. 1995. Microbiology to 10,500 meters in the deep sea. *Ann. Review Microbiology* 49: 777–805.

We now provide a list of selected Web sites that deal with the deep sea. Some of these sites provide highly detailed information. However, Web sites are changing constantly, and occasionally some sites are even cancelled. All of the following sites were in operation as of March 28, 2004:

- Benthos® (ROV): <http://www.benthos.com/>
- JASON project: <http://www.jasonproject.org/>
- NOAA VENTS program: <http://www.pmel.noaa.gov/vents/home.html>
- Woods Hole Oceanographic Institute, Deep Submergence Laboratory: <http://www.dsl.whoi.edu/>

Bacteria

Bioluminescent Bacteria

Kits to culture bioluminescent bacteria are available from both Carolina Biological Supply® and Ward's®. Each kit includes the appropriate medium to successfully culture these interesting bacteria.

The appropriate *Vibrio* (*Photobacterium*) species are also available, albeit at far greater cost, from The America Type Culture Collection (ATCC) (12301 Parklawn Drive, Rockville, MD 20852-1776). (N.B.: The ATCC requires a minimum order amount.)

For more information on *Vibrio fischeri* (ATCC 7744), consult the following sources:

1970. *J. Gen. Microbiol.* 64: 151–169.
1971. *Int. J. Syst. Bact.* 21: 221.
1972. *J. Bact.* 109: 1101–1105.

38 Instructor's Manual for Marine Biology

For more information on *Vibrio harveyi* (ATCC 14126), consult the following sources:

1936. *J. Bact.* 31: 585–592.

1962. *Int. Bull. Bact. Nomen. Taxon.* 12: 77.

1970. *J. Gen. Microbiol.* 64: 151–169.

1980. *Curr. Microbiol.* 4: 127–132.

1980. *Int. J. Syst. Bact.* 30: 316.

Vibrio may be cultured on photobacterium broth.

Anon. 1984. *Difco manual*, 10th ed. Detroit, MI: Difco Laboratories. ISBN 0-9613169-9-3.

Another formulation (photobacterium MPY medium) is reported in the following source:

Atlas, R.M. 1993. *Handbook of microbiological media*. Boca Raton, FL: CRC Press. ISBN 0-8493-2944-2.

Vibrio (family Vibrionaceae) are gram negative, motile, slightly curved to straight rods that grow at about 25°C. Two species of *Vibrio* that have medical importance are *V. cholerae* and *V. parahaemolyticus*; the latter is the etiologic agent of cholera. Both are gut pathogens (enteropathogens) that induce diarrhea, fever, vomiting, and abdominal cramps.

(N.B.: While *Vibrio fischeri* is not considered to be pathogenic, it is a very good idea to treat all bacteria as opportunistic pathogens and to instruct your students accordingly.)

Marine Bacteria

Lepo, J.E., et al. 2003. The effect of amount of crude oil on extent of its biodegradation in open water– and sandy beach– laboratory simulations. *Environ. Techn.* 24: 1291–1304.

Riemann, L., and F. Azam. 2002. Widespread N-acetyl-D-glucosamine uptake among pelagic marine bacteria and its ecological implications. *Appl. Environ. Microbio.* 68: 5554–5563.